

Question 2: AI HW

a) for minimization of simulated annealing: $f(x,y)$

$$\Delta = f_{\text{candidate}} - f_{\text{current}}$$

if $\Delta \leq 0$, we accept otherwise we accept with probability

$$p = e^{-\Delta/T}$$

Booth: $f(x,y) = (x+2y-7)^2 + (2x+y-5)^2$
 $-10 \leq x,y \leq 10$

$$f(0,0) = (0+0-7)^2 + (0+0-5)^2$$

$$= 49 + 25 = 74$$

T: 1 start value

K = 100; iterations

~~Booth~~ =

N.S = 0.5

temp decay = 0.1

Neighborhood size = 0.5 (max one movement size)

Initial temp T = 1

K 100 (iteration before temp change)

Temp decay 0.1 Amount remove after 100 iterations

Booth bounds [-10, 10]

Himmelblau bound [-5, 5]

Griewank bound ~~[-30, 30]~~ (-30, 30)

some random val

eg $-10 \leq x, y \leq 10$

$(5, -2), (-7, 8), (0.6, 3.2)$

own value (test) $f(0,0) = 74$

neighborhood size = 0.5

$\Delta x \in [-0.5, 0.5]$

$\Delta y \in [-0.5, 0.5]$

eg $\Delta x = 0.3$

$\Delta y = 0.4$

$x_{\text{candidate}} = x + \Delta x \Rightarrow 0 + 0.3 = 0.3$

$y_{\text{candidate}} = y + \Delta y = 0 + 0.4 = 0.4$

$(0.3, 0.4) \rightarrow$ in bound $-10 \leq x, y \leq 10$

candidate value:

$f(0.3, 0.4) = 50.81$

$50.81 < 74 \checkmark$

one iteration $\checkmark$ accept

out of 3

Pseudocode:

function SA (prob-function, initial-T, cooling-rate, iterations-per-T):

current-state = random-state-in-bound (bound)

current-value = problem-function (current-state)

best-value = current-value

best-state = current-state

T = initial-T

while T > 0:

for i from 1 to iterations-per-T:

next-state = generate-random-neighbor (current-state)

next-value = problem-function (next-state)

delta-E = next-value - current-value

if delta-E < 0 or Random(0,1) < exp(-delta-E/T):

current-state = next-state

current-value = next-value

if current-value < best-value:

best-state = current-state

best-value = current-value

T = T - cooling-rate

return best-state, best-value

Briefing of the algo code

Goal: Find global minimum of all 3 functions

approach: SA that uses a temp parameter

Normal vs Fine tuned

→ Normal: Cooled too quickly, freezing ^{the} algo before it could reach the space efficiently
in Greinank, it stopped at local minima
 ≈ 0.15

Fine tuning:

- Slow cooling: $T = 5.0$ and reduced decay, 0.05
so algo had more time to explore
- Booth and Himmelblau: their $N_{\text{step}} \rightarrow 0.25$ for
better steps converged to minimum
- Greinank $\rightarrow (-30 - 30)$. Its $N = 1$ to allow
allow our algo to take large search jumps
and reached near approx 0.002